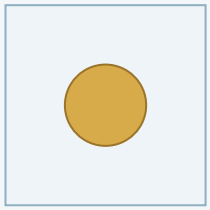
DATASET / GEOMETRY ONLY

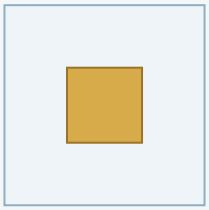
Geometric design space

Four Au shape families. Full 0–320 nm domain, without imposed minimum dimensions.



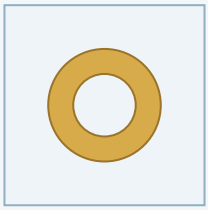
CIRCULAR DISK

D = 130 nm



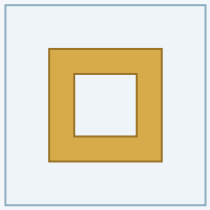
SQUARE

L = 120 nm



CIRCULAR RING

Do = 180, Di = 100



SQUARE RING

Lo = 180, Li = 100

Original dimensions are shown for reference, not as restrictions.

Allowed design parameters

Geometry	Parameters	Allowed values
Disk	Diameter D	$0 \le D \le 320$ nm
Square	Side L	$0 \le L \le 320$ nm
Circular ring	Outer Do, inner Di	$0 \le Di \le Do \le 320$ nm
Square ring	Outer Lo, inner Li	$0 \le Li \le Lo \le 320$ nm

Diameter/side sampling grid

Use n uniformly spaced values from 0 to 320 nm, including both endpoints. The geometry step is

$$\Delta = 320 / (n - 1) \quad \text{nm}$$

Each solid family has n settings. Each ring family has n(n + 1)/2 valid ordered outer/inner pairs, including zero-size and equal-size limits.

$$N_{geometry} = 2n + 2[n(n + 1)/2] = n(n + 3)$$

For every ring, sample all pairs with inner ≤ outer. No inner-diameter, outer-diameter, wall-width, or edge-gap threshold is imposed. At the boundary, some configurations coincide physically and can share one saved result.

14-DAY PLANNING

Diameter step versus dataset size

Complete grids across the four shape families, up to the full 320 nm cell.

Planning assumption: one sequential HFSS worker, 75 minutes per case and 80% effective availability. This runtime is not yet measured for the new geometry campaign. Two existing environmental cases require two results per geometry.

n per size	Step (nm)	Geometries	2 cases	Days
5	80.00	40	80	5.21
6	64.00	54	108	7.03
7	53.33	70	140	9.11
8	45.71	88	176	11.46
9	40.00	108	216	14.06
10	35.56	130	260	16.93
11	32.00	154	308	20.05
17	20.00	340	680	44.27
33	10.00	1,188	2,376	154.69

Recommended uniform step with both cases

n = 8 Δ = 45.71 nm

88 geometry settings × 2 cases = 176 results

Estimated 11.46 days, leaving 2.54 days of buffer.

A 40 nm step needs 216 results and about 14.06 days, already exceeding the limit under the assumed runtime. A 32 nm step requires about 20.05 days. Geometry step size therefore dominates the sample count for the two ring families.

If only one environmental case is retained

The same 14 days permit n = 13 (Δ = 26.67 nm): 208 unique grid settings and about 13.54 days at 75 minutes per result.

Runtime sensitivity: with two cases, n = 8 requires 9.17 days at 60 min/case, 11.46 at 75 min, and 13.75 at 90 min. Confirm the average runtime before fixing the step.

Counts include boundary and equivalent limiting settings; deduplication can only reduce work.